Bevel Gearing

Fundamental Contact Stress Equation

$$s_c = \sigma_c = C_p\left(\frac{W^t}{Fd_PI}K_oK_vK_mC_sC_{xc}\right)^{1/2} \qquad \text{(U.S. customary units)}$$

$$\sigma_H = Z_E\left(\frac{1000W^t}{bdZ_1}K_AK_vK_{H\beta}Z_xZ_{xc}\right)^{1/2} \qquad \text{(SI units)} \qquad (15\text{–}1)$$

Permissible Contact Stress Number (Strength) Equation

$$s_{wc} = (\sigma_c)_{\text{all}} = \frac{s_{ac}C_LC_H}{S_HK_TC_R} \qquad \text{(U.S. customary units)}$$

$$\sigma_{HP} = \frac{\sigma_{H\,\text{lim}}Z_{NT}Z_W}{S_HK_\theta Z_Z} \qquad \text{(SI units)} \qquad (15\text{–}2)$$

Bending Stress

$$s_t = \frac{W^t}{F}P_dK_oK_v\frac{K_sK_m}{K_xJ} \qquad \text{(U.S. customary units)}$$

$$\sigma_F = \frac{1000W^t}{b}\frac{K_AK_v}{m_{et}}\frac{Y_xK_{H\beta}}{Y_\beta Y_J} \qquad \text{(SI units)} \qquad (15\text{–}3)$$

Permissible Bending Stress Equation

$$s_{wt} = \frac{s_{at}K_L}{S_FK_TK_R} \qquad \text{(U.S. customary units)}$$

$$\sigma_{FP} = \frac{\sigma_{F\,\text{lim}}Y_{NT}}{S_FK_\theta Y_z} \qquad \text{(SI units)} \qquad (15\text{–}4)$$

AGMA Equation Factors

$$K_v = \left(\frac{A + \sqrt{v_t}}{A}\right)^B \qquad \text{(U.S. customary units)}$$

$$K_v = \left(\frac{A + \sqrt{200v_{et}}}{A}\right)^B \qquad \text{(SI units)} \qquad (15\text{–}5)$$

Where

$$A = 50 + 56(1 - B)$$
$$B = 0.25(12 - Q_v)^{2/3} \qquad (15\text{–}6)$$

$$v_t = \pi d_P n_P / 12 \qquad \text{(U.S. customary units)}$$
$$v_{et} = 5.236(10^{-5}) d_1 n_1 \qquad \text{(SI units)} \qquad (15\text{–}7)$$

$$v_{t\,\max} = [A + (Q_v - 3)]^2 \qquad \text{(U.S. customary units)}$$
$$v_{et\,\max} = \frac{[A + (Q_v - 3)]^2}{200} \qquad \text{(SI units)} \qquad (15\text{–}8)$$

Size Factor for Pitting Resistance Cs (*Zx*)

$$C_s = \begin{cases} 0.5 & F < 0.5 \text{ in} \\ 0.125F + 0.4375 & 0.5 \le F \le 4.5 \text{ in} \\ 1 & F > 4.5 \text{ in} \end{cases} \qquad \text{(U.S. customary units)} \qquad (15\text{–}9)$$

$$Z_x = \begin{cases} 0.5 & b < 12.7 \text{ mm} \\ 0.004\,92b + 0.4375 & 12.7 \le b \le 114.3 \text{ mm} \\ 1 & b > 114.3 \text{ mm} \end{cases} \qquad \text{(SI units)}$$

Size Factor for Bending Ks (*Yx*)

$$K_S = \begin{cases} 0.4867 + 0.2132/P_d & 0.5 \le P_d \le 16 \text{ teeth/in} \\ 0.5 & P_d > 16 \text{ teeth/in} \end{cases} \qquad \text{(U.S. customary units)} \qquad (15\text{–}10)$$

$$Y_x = \begin{cases} 0.5 & m_{et} < 1.6 \text{ mm} \\ 0.4867 + 0.008\,339 m_{et} & 1.6 \le m_{et} \le 50 \text{ mm} \end{cases} \qquad \text{(SI units)}$$

Load-Distribution Factor Km ($K_{H\beta}$)

$$K_m = K_{mb} + 0.0036F^2 \qquad \text{(U.S. customary units)}$$
$$K_{H\beta} = K_{mb} + 5.6(10^{-6})b^2 \qquad \text{(SI units)} \qquad (15\text{–}11)$$

where

$$K_{mb} = \begin{cases} 1.00 & \text{both members straddle-mounted} \\ 1.10 & \text{one member straddle-mounted} \\ 1.25 & \text{neither member straddle-mounted} \end{cases}$$

Crowning Factor for Pitting Cxc (Zxc)

$$C_{xc} = Z_{xc} = \begin{cases} 1.5 & \text{properly crowned teeth} \\ 2.0 & \text{or larger uncrowned teeth} \end{cases} \qquad (15\text{–}12)$$

Lengthwise Curvature Factor for Bending Strength Kx (Y_β)

For straight-bevel gears,

$$K_x = Y_\beta = 1 \qquad (15\text{–}13)$$

Stress-Cycle Factor for Pitting Resistance C_L (Z_{NT})

$$C_L = \begin{cases} 2 & 10^3 \le N_L < 10^4 \\ 3.4822 N_L^{-0.0602} & 10^4 \le N_L \le 10^{10} \end{cases}$$

$$Z_{NT} = \begin{cases} 2 & 10^3 \le n_L < 10^4 \\ 3.4822 n_L^{-0.0602} & 10^4 \le n_L \le 10^{10} \end{cases} \qquad (15\text{–}14)$$

Stress-Cycle Factor for Bending Strength K_L (Y_{NT})

$$K_L = \begin{cases} 2.7 & 10^2 \le N_L < 10^3 \\ 6.1514 N_L^{-0.1192} & 10^3 \le N_L < 3(10^6) \\ 1.683 N_L^{-0.0323} & 3(10^6) \le N_L \le 10^{10} \quad \text{critical} \\ 1.3558 N_L^{-0.0178} & 3(10^6) \le N_L \le 10^{10} \quad \text{general} \end{cases}$$

$$Y_{NT} = \begin{cases} 2.7 & 10^2 \le n_L < 10^3 \\ 6.1514 n_L^{-0.1192} & 10^3 \le n_L < 3(10^6) \\ 1.683 n_L^{-0.0323} & 3(10^6) \le n_L \le 10^{10} \quad \text{critical} \\ 1.3558 n_L^{-0.0178} & 3(10^6) \le n_L \le 10^{10} \quad \text{general} \end{cases} \qquad (15\text{–}15)$$

Hardness-Ratio Factor C_H (Z_W)

$$C_H = 1 + B_1(N/n - 1) \qquad B_1 = 0.008\,98(H_{BP}/H_{BG}) - 0.008\,29$$

$$Z_W = 1 + B_1(z_2/z_1 - 1) \qquad B_1 = 0.008\,98(H_{B1}/H_{B2}) - 0.008\,29 \qquad (15\text{–}16)$$

The preceding equations are valid when $1.2 \le H_{BP}/H_{BG} \le 1.7$ ($1.2 \le H_{B1}/H_{B2} \le 1.7$).

$$C_H = 1 + B_2(450 - H_{BG}) \qquad B_2 = 0.000\,75 \exp(-0.0122 f_P)$$

$$Z_W = 1 + B_2(450 - H_{B2}) \qquad B_2 = 0.000\,75 \exp(-0.52 R_{a1}) \qquad (15\text{–}17)$$

Temperature Factor K_T (K_θ)

$$K_T = \begin{cases} 1 & 32°\text{F} \le t \le 250°\text{F} \\ (460 + t)/710 & t > 250°\text{F} \end{cases}$$

$$K_\theta = \begin{cases} 1 & 0°\text{C} \le \theta \le 120°\text{C} \\ (273 + \theta)/393 & \theta > 120°\text{C} \end{cases} \qquad (15\text{–}18)$$

Reliability Factors C_R (Z_Z) and K_R (Y_Z)

$$C_R = \sqrt{K_R} \text{ and } Z_Z = \sqrt{Y_Z}.$$

$$Y_Z = K_R = \begin{cases} 0.50 - 0.25 \log(1 - R) & 0.99 \le R \le 0.999 \quad (15\text{–}19) \\ 0.70 - 0.15 \log(1 - R) & 0.90 \le R < 0.99 \quad (15\text{–}20) \end{cases}$$

The reliability of the stress (fatigue) numbers allowable in Tables 15–4, 15–5, 15–6, and 15–7 is 0.99.

Elastic Coefficient for Pitting Resistance Cp (Z_E)

$$C_p = \sqrt{\frac{1}{\pi[(1 - \nu_P^2)/E_P + (1 - \nu_G^2)/E_G]}}$$

$$Z_E = \sqrt{\frac{1}{\pi[(1 - \nu_1^2)/E_1 + (1 - \nu_2^2)/E_2]}} \qquad (15\text{–}21)$$

where C_p = elastic coefficient, 2290 $\sqrt{\text{psi}}$ for steel

Z_E = elastic coefficient, 190 $\sqrt{\text{N/mm}^2}$ for steel

E_P and E_G = Young's moduli for pinion and gear respectively, psi

E_1 and E_2 = Young's moduli for pinion and gear respectively, N/mm^2

Allowable Contact Stress

$$\begin{aligned} s_{ac} &= 341H_B + 23\,620 \text{ psi} && \text{grade 1} \\ \sigma_{H\,\text{lim}} &= 2.35H_B + 162.89 \text{ MPa} && \text{grade 1} \\ s_{ac} &= 363.6H_B + 29\,560 \text{ psi} && \text{grade 2} \\ \sigma_{H\,\text{lim}} &= 2.51H_B + 203.86 \text{ MPa} && \text{grade 2} \end{aligned} \qquad (15\text{–}22)$$

Allowable Bending Stress Numbers

$$\begin{aligned} s_{at} &= 44H_B + 2100 \text{ psi} && \text{grade 1} \\ \sigma_{F\,\lim} &= 0.30H_B + 14.48 \text{ MPa} && \text{grade 1} \\ s_{at} &= 48H_B + 5980 \text{ psi} && \text{grade 2} \\ \sigma_{H\,\lim} &= 0.33H_B + 41.24 \text{ MPa} && \text{grade 2} \end{aligned} \qquad (15\text{–}23)$$

Design of a Straight-Bevel Gear Mesh

$$F = \min(0.3A_0, 10/P_d) \qquad (15\text{–}24)$$

where A_0 is the cone distance (see Fig. 13–20, p. 682), given by

$$A_0 = \frac{d_P}{2\sin\gamma} = \frac{d_G}{2\sin\Gamma} \qquad (15\text{–}25)$$

Worm Gearing—AGMA Equation

$$\sum = \psi_P \pm \psi_G \qquad (15\text{–}26)$$

$$\frac{C^{0.875}}{3} \le d \le \frac{C^{0.875}}{1.6} \qquad (15\text{–}27)$$

$$(W^t)_{\text{all}} = C_s D_m^{0.8} F_e C_m C_v \qquad (15\text{–}28)$$

F_e *must not exceed* $0.67d_m$

$$W_f = \frac{f W^t}{\cos\lambda \cos\phi_n} \qquad (15\text{–}29)$$

$$V_s = \frac{\pi n_W d_m}{12\cos\lambda} \qquad (15\text{–}30)$$

$$T_G = \frac{W^t D_m}{2} \qquad (15\text{–}31)$$

$$C_s = 720 + 10.37C^3 \qquad C \le 3 \text{ in} \qquad (15\text{–}32)$$

For sand-cast gears,

$$C_s = 720 + 10.37C^3 \qquad C \le 3 \text{ in} \qquad (15\text{–}32)$$

For chilled-cast gears,

$$C_s = \begin{cases} 1000 & C > 3 \quad D_m \le 2.5 \text{ in} \\ 1190 - 477 \log D_m & C > 3 \quad D_m > 2.5 \text{ in} \end{cases} \tag{15–33}$$

For centrifugally cast gears,

$$C_s = \begin{cases} 1000 & C > 3 \quad D_m \le 25 \text{ in} \\ 1251 - 180 \log D_m & C > 3 \quad D_m > 25 \text{ in} \end{cases} \tag{15–35}$$

$$C_m = \begin{cases} 0.02\sqrt{-m_G^2 + 40m_G - 76} + 0.46 & 3 < m_G \le 20 \\ 0.0107\sqrt{-m_G^2 + 56m_G + 5145} & 20 < m_G \le 76 \\ 1.1483 - 0.006\,58m_G & m_G > 76 \end{cases} \tag{15–36}$$

$$C_v = \begin{cases} 0.659 \exp(-0.0011V_s) & V_s < 700 \text{ ft/min} \\ 13.31\, V_s^{-0.571} & 700 \le V_s < 3000 \text{ ft/min} \\ 65.52\, V_s^{-0.774} & V_s > 3000 \text{ ft/min} \end{cases} \tag{15–37}$$

$$f = \begin{cases} 0.15 & V_s = 0 \\ 0.124 \exp(-0.074V_s^{0.645}) & 0 < V_s \le 10 \text{ ft/min} \\ 0.103 \exp(-0.110V_s^{0.450}) + 0.012 & V_s > 10 \text{ ft/min} \end{cases} \tag{15–38}$$

$$a = \frac{p_x}{\pi} = 0.3183p_x \tag{15–39}$$

$$b = \frac{1.157p_x}{\pi} = 0.3683p_x \tag{15–40}$$

$$h_t = \begin{cases} \dfrac{2.157p_x}{\pi} = 0.6866p_x & p_x \ge 0.16 \text{ in} \\ \dfrac{2.200p_x}{\pi} + 0.002 = 0.7003p_x + 0.002 & p_x < 0.16 \text{ in} \end{cases} \tag{15–41}$$

$$d_o = d + 2a \tag{15–42}$$

$$d_r = d - 2b \tag{15–43}$$

$$D_t = D + 2a \tag{15–44}$$

$$D_r = D - 2b \tag{15–45}$$

$$c = b - a \tag{15–46}$$

$$(F_W)_{\max} = 2\sqrt{\left(\frac{D_t}{2}\right)^2 - \left(\frac{D}{2} - a\right)^2} = 2\sqrt{2Da} \tag{15–47}$$

$$F_G = \begin{cases} 2d_m/3 & p_x > 0.16 \text{ in} \\ 1.125\sqrt{(d_o + 2c)^2 - (d_o - 4a)^2} & p_x \le 0.16 \text{ in} \end{cases} \tag{15–48}$$

$$H_{\text{loss}} = 33\,000(1 - e)H_{\text{in}} \tag{15–49}$$

$$\hbar_{\text{CR}} = \begin{cases} \dfrac{n_W}{6494} + 0.13 & \text{no fan on worm shaft} \\ \dfrac{n_W}{3939} + 0.13 & \text{fan on worm shaft} \end{cases} \tag{15–50}$$

$$t_s = t_a + \frac{H_{\text{loss}}}{\hbar_{\text{CR}} A} = \frac{33\,000(1 - e)(H)_{\text{in}}}{\hbar_{\text{CR}} A} + t_a \tag{15–51}$$

$$A_{\min} = 43.20C^{1.7} \tag{15–52}$$

$$\sigma_a = \frac{W_G^t}{p_n F_e y} \tag{15–53}$$

where $p_n = p_x \cos \lambda$ and y is the Lewis form factor related to circular pitch. For $\phi_n = 14.5°$, $y = 0.100$; $\phi_n = 20°$, $y = 0.125$; $\phi_n = 25°$, $y = 0.150$; $\phi_n = 30°$, $y = 0.175$.

Worm-Gear Analysis

$$e_W = \frac{\cos \phi_n - f \tan \lambda}{\cos \phi_n + f \cot \lambda} \tag{15–54}$$

$$e_G = \frac{\cos \phi_n - f \cot \lambda}{\cos \phi_n + f \tan \lambda} \tag{15–55}$$

$$f_{\text{stat}} < \cos \phi_n \tan \lambda \tag{15–56}$$

$$W_W^t = W_G^t \frac{\cos \phi_n \sin \lambda + f \cos \lambda}{\cos \phi_n \cos \lambda - f \sin \lambda} \tag{15–57}$$

$$W_G^t = \frac{33\,000 n_d H_0 K_a}{V_G e} \tag{15–58}$$

$$H_W = \frac{W_W^t V_W}{33\,000} = \frac{\pi d_W n_W W_W^t}{12(33\,000)} \tag{15–59}$$

$$H_G = \frac{W_G^t V_G}{33\,000} = \frac{\pi d_G n_G W_G^t}{12(33\,000)} \tag{15–60}$$

$$W_f = \frac{f W_G^t}{f \sin \lambda - \cos \phi_n \cos \lambda} \tag{15–61}$$

$$V_s = \frac{\pi d n_W}{12 \cos \lambda} \tag{15–62}$$

$$H_f = \frac{|W_f| V_s}{33\,000} \text{ hp} \tag{15–63}$$

Designing a Worm-Gear Mesh

Typical pitches are $\frac{1}{4}, \frac{5}{16}, \frac{3}{8}, \frac{1}{2}, \frac{3}{4}, 1, \frac{5}{4}, \frac{6}{4}, \frac{7}{4}$, and 2.

$$N_G = m_G N_W \qquad P_t = \frac{\pi}{p_x} \qquad D = \frac{N_G}{P_t}$$

$$(d)_{\text{lo}} = C^{0.875}/3 \qquad (d)_{\text{hi}} = C^{0.875}/1.6$$

$$C = (d_1 + D)/2.$$

$$d_2 = 2C - D$$

Buckingham Wear Load

$$(W_G^t)_{\text{all}} = K_w d_G F_e \tag{15–64}$$

Table 15–2

Overload Factors K_o (K_A)

Source: ANSI/AGMA 2003-B97.

Character of Prime Mover	Character of Load on Driven Machine			
	Uniform	Light Shock	Medium Shock	Heavy Shock
Uniform	1.00	1.25	1.50	1.75 or higher
Light shock	1.10	1.35	1.60	1.85 or higher
Medium shock	1.25	1.50	1.75	2.00 or higher
Heavy shock	1.50	1.75	2.00	2.25 or higher

Note: This table is for speed-decreasing drives. For speed-increasing drives, add $0.01(N/n)^2$ or $0.01(z_2/z_1)^2$ to the above factors.

Figure 15–5

Dynamic factor K_v. *(Source: ANSI/AGMA 2003-B97.)*

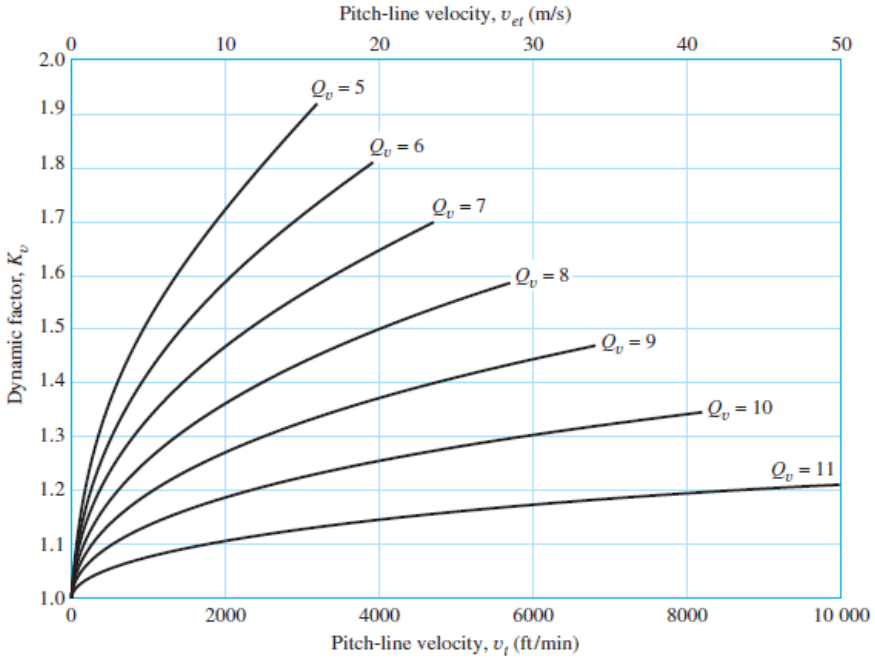


Figure 15–6

Contact geometry factor I (Z_I) for coniflex straight-bevel gears with a 20° normal pressure angle and a 90° shaft angle. *(Source: ANSI/AGMA 2003-B97.)*

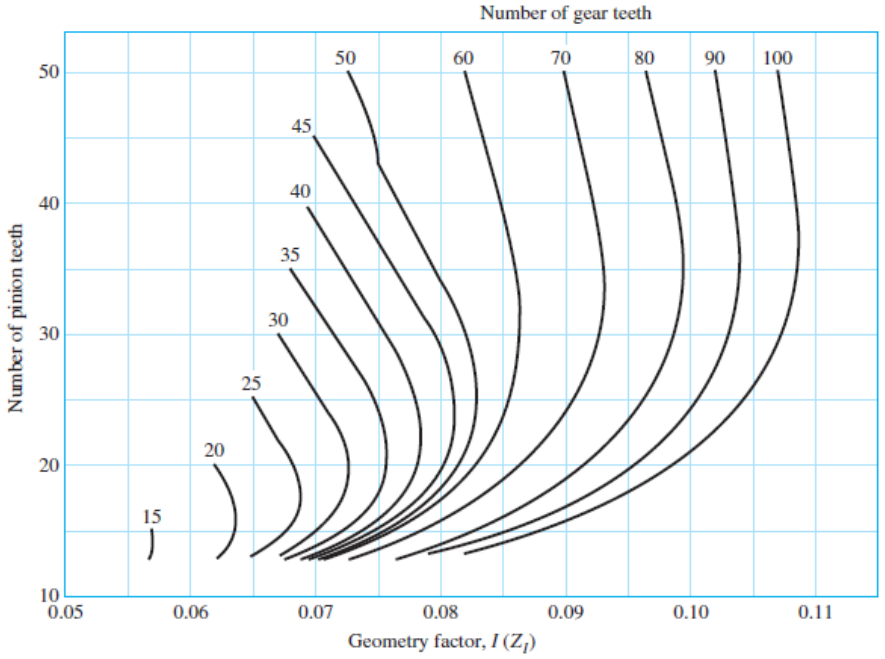
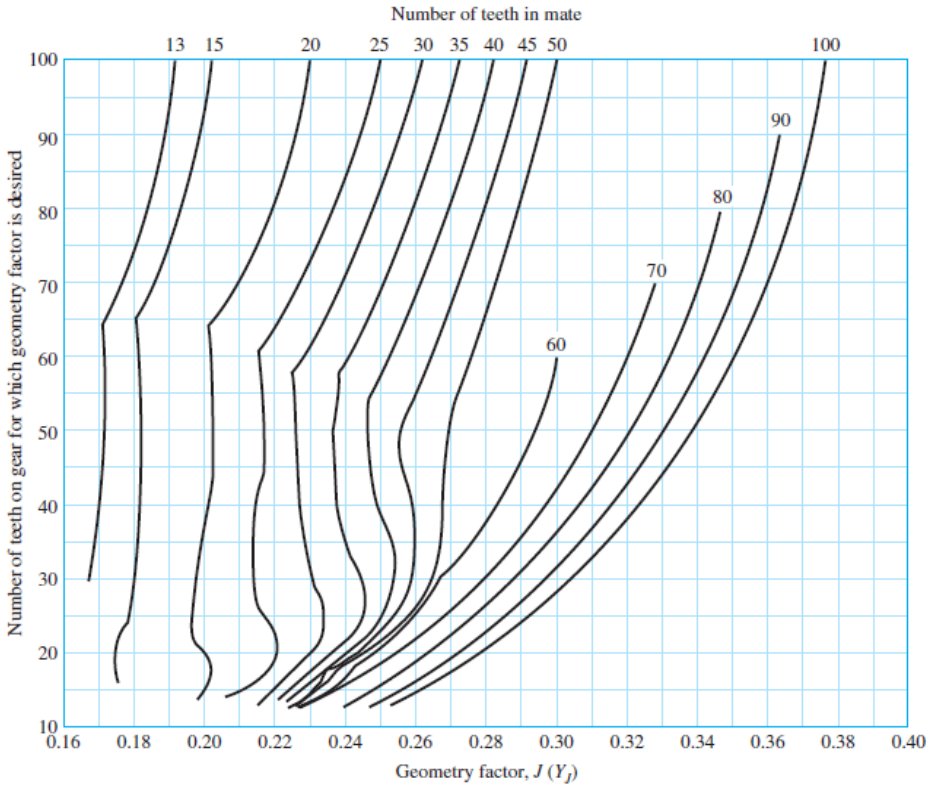


Figure 15–7

Bending factor J (Y_J) for coniflex straight-bevel gears with a 20° normal pressure angle and 90° shaft angle. *(Source: ANSI/AGMA 2003-B97.)*



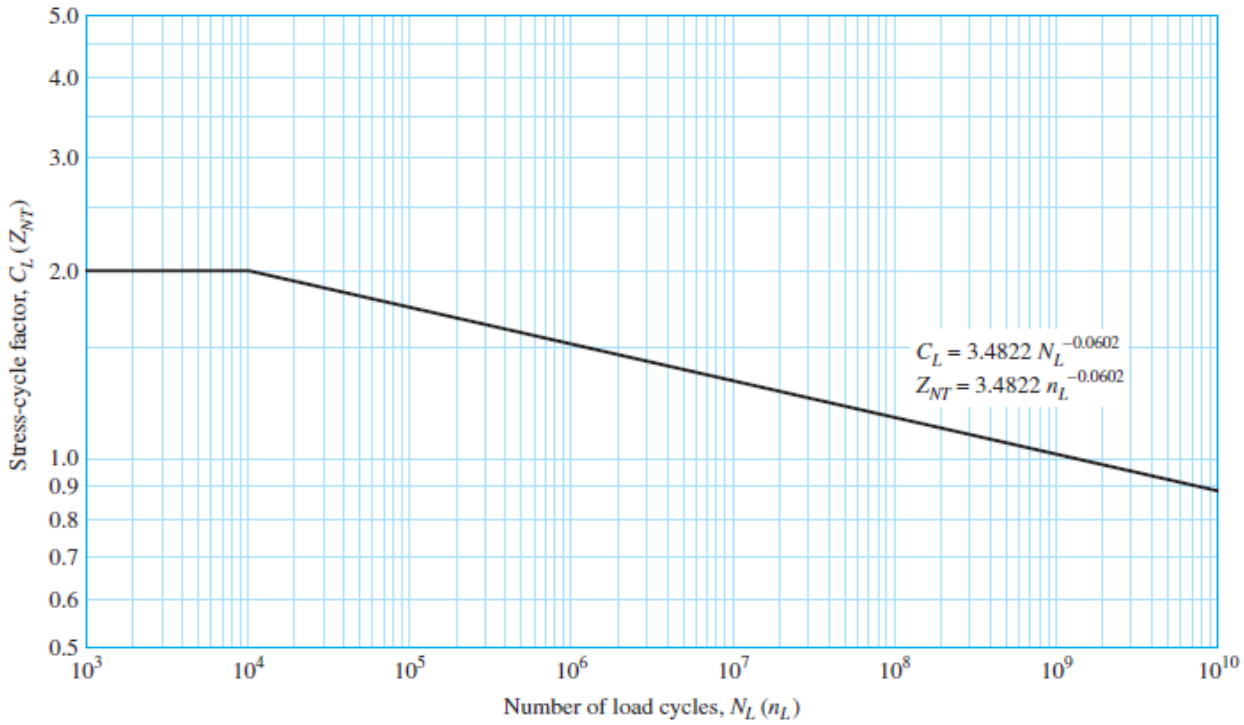


Figure 15–8

Contact stress-cycle factor for pitting resistance C_L (Z_{NT}) for carburized case-hardened steel bevel gears. *(Source: ANSI/AGMA 2003-B97.)*

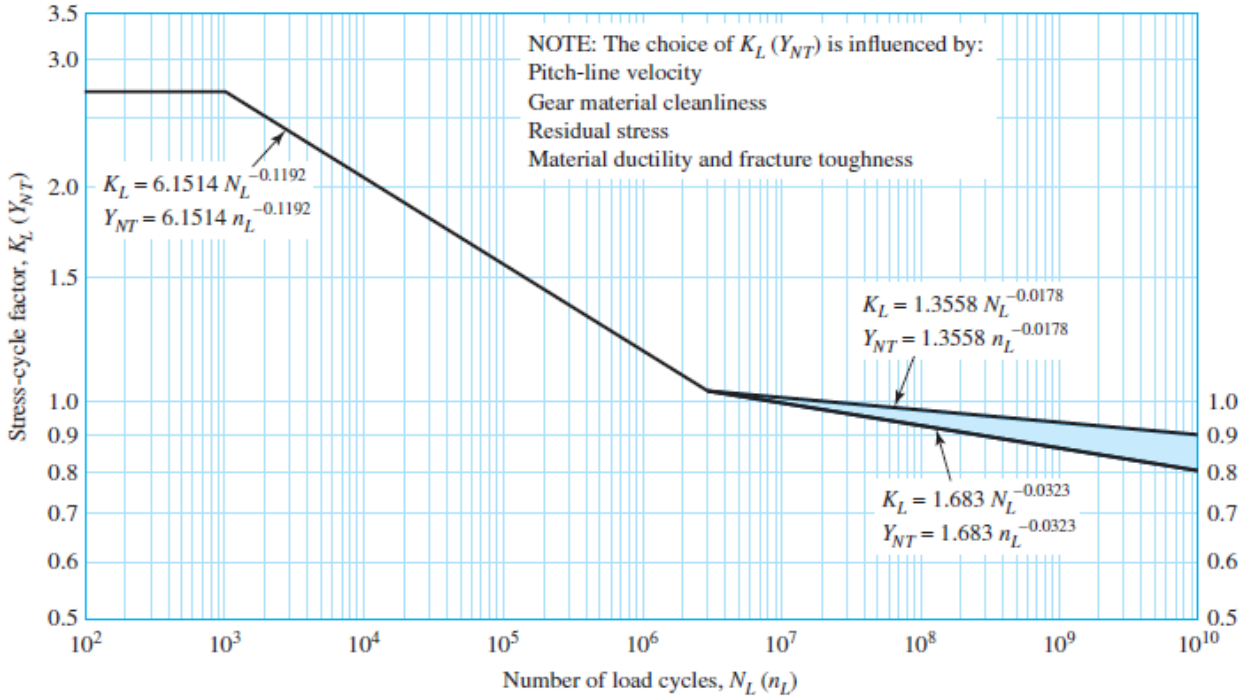


Figure 15–9

Stress-cycle factor for bending strength K_L (Y_{NT}) for carburized case-hardened steel bevel gears. *(Source: ANSI/AGMA 2003-B97.)*

Figure 15–10

Hardness-ratio factor C_H (Z_W) for through-hardened pinion and gear. *(Source: ANSI/AGMA 2003-B97.)*

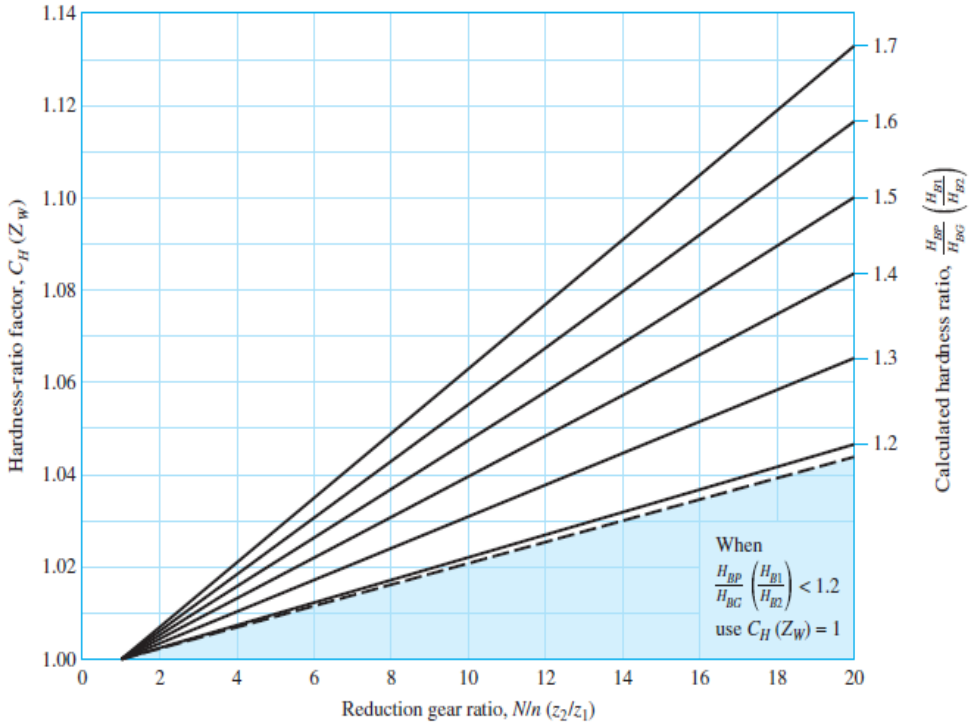


Figure 15–11

Hardness-ratio factor C_H (Z_W) for surface-hardened pinions. *(Source: ANSI/AGMA 2003-B97.)*

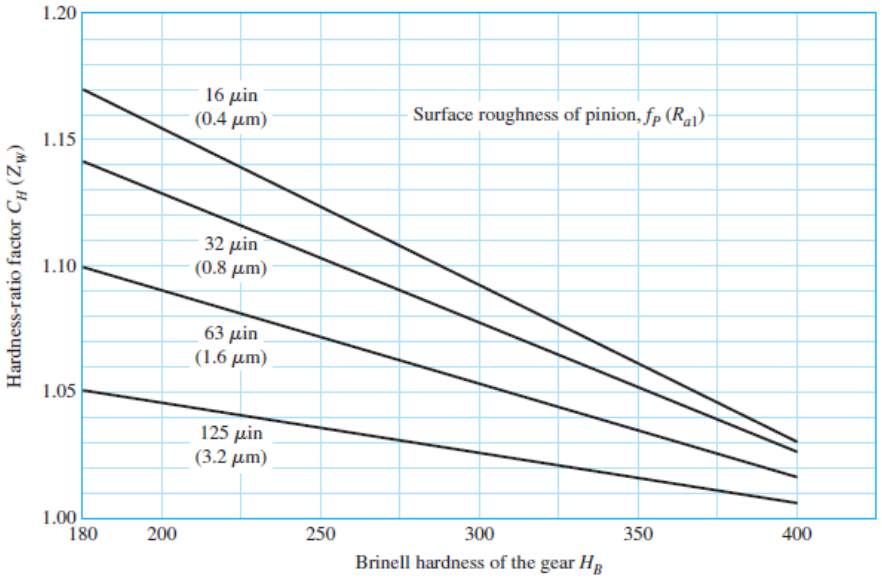


Table 15–3

Reliability Factors

Source: ANSI/AGMA 2003-B97.

Requirements of Application	Reliability Factors for Steel* C_R (Z_Z)	K_R (Y_Z)†
Fewer than one failure in 10 000	1.22	1.50
Fewer than one failure in 1000	1.12	1.25
Fewer than one failure in 100	1.00	1.00
Fewer than one failure in 10	0.92	0.85‡
Fewer than one failure in 2	0.84	0.70§

Table 15–4

Allowable Contact Stress Number for Steel Gears, s_{ac} ($\sigma_{H\,\mathrm{lim}}$) *Source: ANSI/AGMA 2003-B97.*

Material Designation	Heat Treatment	Minimum Surface* Hardness	Allowable Contact Stress Number, s_{ac} ($\sigma_{H\,\mathrm{lim}}$) lbf/in² (N/mm²) Grade 1†	Grade 2†	Grade 3†
Steel	Through-hardened‡	Fig. 15–12	Fig. 15–12	Fig. 15–12	
	Flame or induction hardened§	50 HRC	175 000 (1210)	190 000 (1310)	
	Carburized and case hardened§	2003-B97 Table 8	200 000 (1380)	225 000 (1550)	250 000 (1720)
AISI 4140	Nitrided§	84.5 HR15N		145 000 (1000)	
Nitralloy 135M	Nitrided§	90.0 HR15N		160 000 (1100)	

Table 15–5

Allowable Contact Stress Number for Iron Gears, s_{ac} ($\sigma_{H\,\lim}$) *Source: ANSI/AGMA 2003-B97.*

Material	Material Designation ASTM	Material Designation ISO	Heat Treatment	Typical Minimum Surface Hardness	Allowable Contact Stress Number, s_{ac} ($\sigma_{H\,\lim}$) lbf/in^2 (N/mm^2)
Cast iron	ASTM A48	ISO/DR 185			
	Class 30	Grade 200	As cast	175 HB	50 000 (345)
	Class 40	Grade 300	As cast	200 HB	65 000 (450)
Ductile (nodular) iron	ASTM A536	ISO/DIS 1083			
	Grade 80-55-06	Grade 600-370-03	Quenched	180 HB	94 000 (650)
	Grade 120-90-02	Grade 800-480-02	and tempered	300 HB	135 000 (930)

Table 15–6

Allowable Bending Stress Numbers for Steel Gears, s_{at} ($\sigma_{F\,\lim}$) *Source: ANSI/AGMA 2003-B97.*

Material Designation	Heat Treatment	Minimum Surface Hardness	Bending Stress Number (Allowable), s_{at} ($\sigma_{F\,\lim}$) lbf/in^2 (N/mm^2) Grade 1*	Grade 2*	Grade 3*
Steel	Through-hardened	Fig. 15–13	Fig. 15–13	Fig. 15–13	
	Flame or induction hardened				
	Unhardened roots	50 HRC	15 000 (85)	13 500 (95)	
	Hardened roots		22 500 (154)		
	Carburized and case hardened†	2003-B97 Table 8	30 000 (205)	35 000 (240)	40 000 (275)
AISI 4140	Nitrided†,‡	84.5 HR15N		22 000 (150)	
Nitralloy 135M	Nitrided†,‡	90.0 HR15N		24 000 (165)	

Table 15–7

Allowable Bending Stress Number for Iron Gears, s_{at} ($\sigma_{F\,\lim}$) *Source: ANSI/AGMA 2003-B97.*

Material	Material Designation ASTM	Material Designation ISO	Heat Treatment	Typical Minimum Surface Hardness	Bending Stress Number (Allowable), s_{at} ($\sigma_{F\,\lim}$) lbf/in^2 (N/mm^2)
Cast iron	ASTM A48	ISO/DR 185			
	Class 30	Grade 200	As cast	175 HB	4500 (30)
	Class 40	Grade 300	As cast	200 HB	6500 (45)
Ductile (nodular) iron	ASTM A536	ISO/DIS 1083			
	Grade 80-55-06	Grade 600-370-03	Quenched	180 HB	10 000 (70)
	Grade 120-90-02	Grade 800-480-02	and tempered	300 HB	13 500 (95)

Figure 15–12

Allowable contact stress number for through-hardened steel gears, $s_{ac}(\sigma_{H\ \mathrm{lim}})$. *(Source: ANSI/AGMA 2003-B97.)*

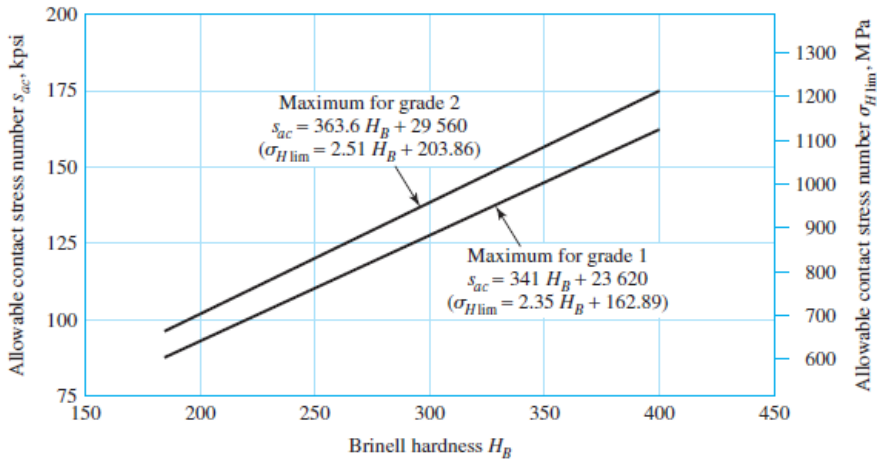


Figure 15–13

Allowable bending stress number for through-hardened steel gears, $s_{at}(\sigma_{F\ \mathrm{lim}})$. *(Source: ANSI/AGMA 2003-B97.)*

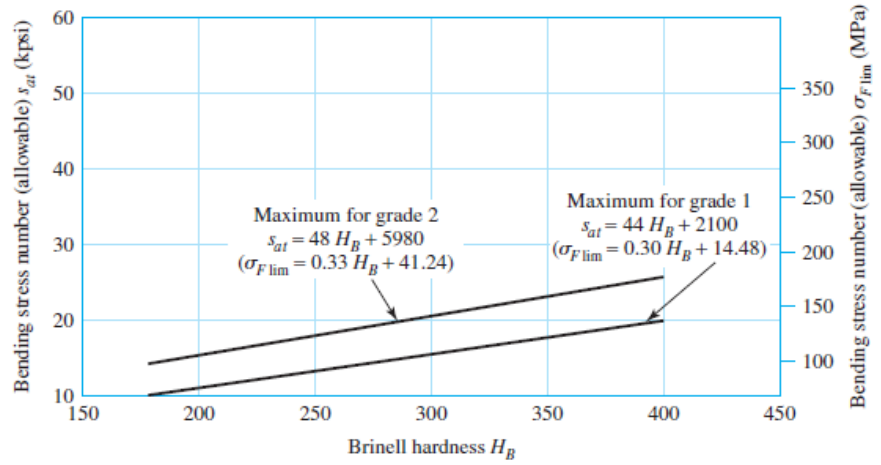


Figure 15–14

"Road map" summary of principal straight-bevel gear wear equations and their parameters.

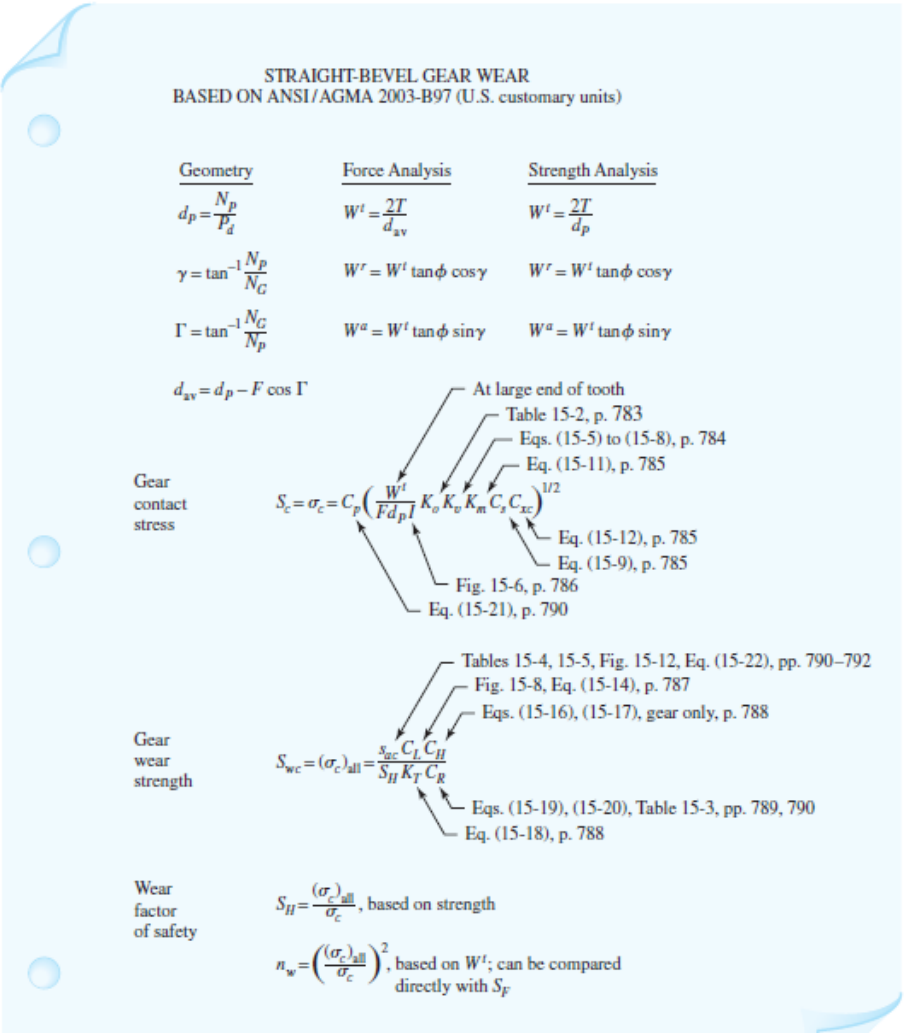
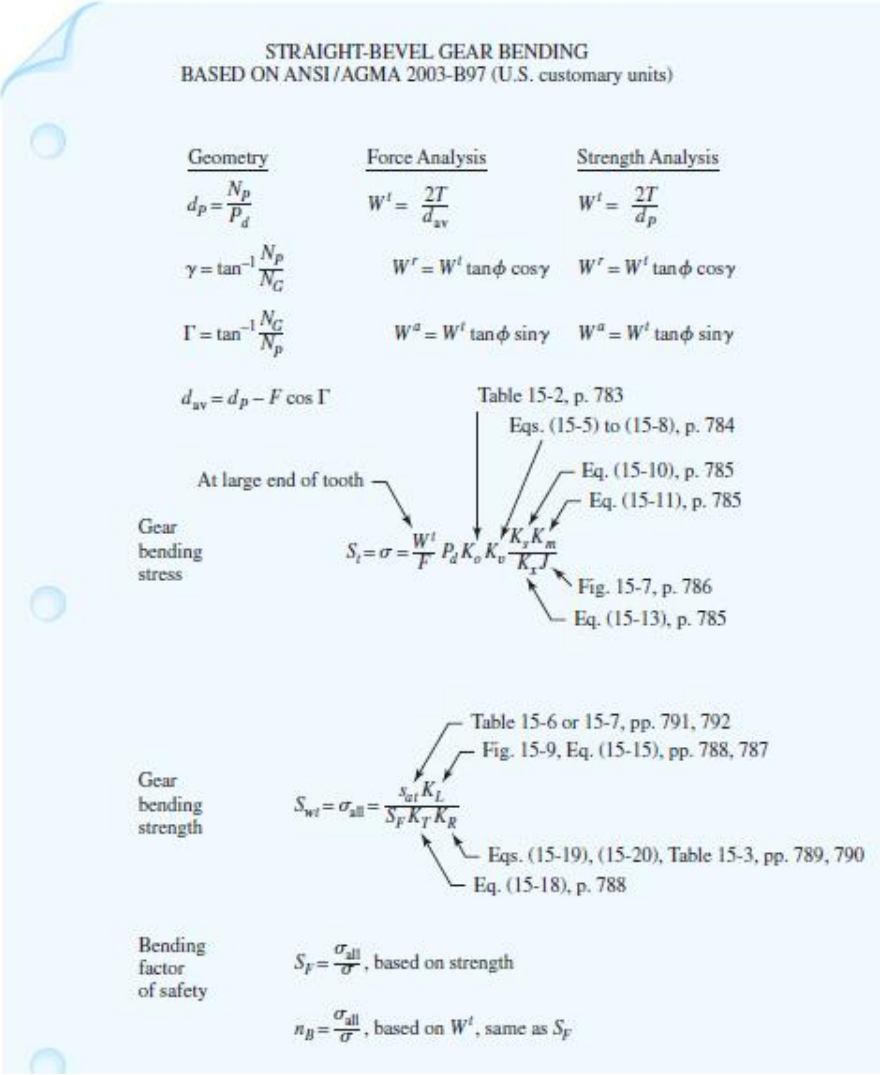


Figure 15–15

"Road map" summary of principal straight-bevel gear bending equations and their parameters.



The conversion from HRC to H_B is

HRC	42	40	38	36	34	32	30	28	26	24	22	20	18	16	14	12	10
H_B	388	375	352	331	321	301	285	269	259	248	235	223	217	207	199	192	187

Table 15–8

Cylindrical Worm Dimensions Common to Both Worm and Gear*

Quantity	Symbol	ϕ_n 14.5° $N_W \le 2$	ϕ_n 20° $N_W \le 2$	ϕ_n 25° $N_W > 2$
Addendum	a	$0.3183p_x$	$0.3183p_x$	$0.286p_x$
Dedendum	b	$0.3683p_x$	$0.3683p_x$	$0.349p_x$
Whole depth	h_t	$0.6866p_x$	$0.6866p_x$	$0.635p_x$

*The table entries are for a tangential diametral pitch of the gear of $P_t = 1$.

Table 15–9

Largest Lead Angle Associated with a Normal Pressure Angle ϕ_n for Worm Gearing

ϕ_n	Maximum Lead Angle λ_{max}
14.5°	16°
20°	25°
25°	35°
30°	45°

Table 15–10

Minimum Number of Gear Teeth for Normal Pressure Angle ϕ_n

ϕ_n	$(N_G)_{min}$
14.5	40
17.5	27
20	21
22.5	17
25	14
27.5	12
30	10

Table 15–11

Wear Factor K_w for Worm Gearing

Source: Earle Buckingham, *Design of Worm and Spiral Gears,* Industrial Press, New York, 1981.

Material		Thread Angle ϕ_n			
Worm	Gear	$14\frac{1}{2}°$	20°	25°	30°
Hardened steel*	Chilled bronze	90	125	150	180
Hardened steel*	Bronze	60	80	100	120
Steel, 250 BHN (min.)	Bronze	36	50	60	72
High-test cast iron	Bronze	80	115	140	165
Gray iron†	Aluminum	10	12	15	18
High-test cast iron	Gray iron	90	125	150	180
High-test cast iron	Cast steel	22	31	37	45
High-test cast iron	High-test cast iron	135	185	225	270
Steel 250 BHN (min.)	Laminated phenolic	47	64	80	95
Gray iron	Laminated phenolic	70	96	120	140

*Over 500 BHN surface.

†For steel worms, multiply given values by 0.6.